

2023 Ch H2 Q7

Section: Chemistry in Society

Topic: Water Chemistry / Environmental Chemistry

Question Summary

Tap water contains dissolved chemicals.

This question covers:

(a) Hard water, soapless detergents, and titration with EDTA to measure Ca^{2+} concentration.

(b) Determining MnO_4^- concentration using colorimetry and calibration graph.

(c) Polarity of trichloromethane vs tetrachloromethane; enthalpy calculation for reaction with HF.

Worked Solution

According to the SQA Marking Instructions for 2023 H2 Q7:

(a)(i) Water with high levels of dissolved $\text{Ca}^{2+}/\text{Mg}^{2+}$ salts is called hard water.

(a)(ii) The ionic part of the detergent molecule is the hydrophilic head group (sulfonate group).

(a)(iii) Average titre = $9.3 \text{ cm}^3 = 0.0093 \text{ L}$.
Moles EDTA = $0.0093 \times 0.0045 = 4.19 \times 10^{-5} \text{ mol}$. Reaction 1:1 $\rightarrow$ moles $\text{Ca}^{2+} = 4.19 \times 10^{-5}$. Concentration in $50.0 \text{ cm}^3 = 4.19 \times 10^{-5} \div 0.050 = 8.38 \times 10^{-4} \text{ mol L}^{-1}$.

(b) From the calibration graph: absorbance 0.08 corresponds to about $2.5 \text{ mg L}^{-1} \text{ MnO}_4^-$.

(c)(i) Trichloromethane (CHCl_3) is polar because the C-H bond is not cancelled by the C-Cl bonds; the dipoles do not fully cancel, giving a net polarity.

Tetrachloromethane (CCl_4) is symmetrical, so dipoles cancel $\rightarrow$ non-polar.

(c)(ii) Bonds broken: 1 C-H (412), 3 C-Cl ($3 \times 338 = 1014$), 1 H-F (565). Total = 1991 kJ.

Bonds formed: 1 C-F (484), 3 H-Cl ($3 \times 431 = 1293$). Total = 1777 kJ.

$\Delta H = 1991 - 1777 = +214 \text{ kJ mol}^{-1}$

(endothermic).

Final Answer

(a)(i) Hard water.

**(a)(ii) Hydrophilic ionic head
(sulfonate group).**

(a)(iii) $[\text{Ca}^{2+}] = 8.38 \times 10^{-4} \text{ mol L}^{-1}$.

(b) $[\text{MnO}_4^-] \approx 2.5 \text{ mg L}^{-1}$.

**(c)(i) CHCl_3 polar; CCl_4 non-polar
(symmetry).**

(c)(ii) $\Delta H = +214 \text{ kJ mol}^{-1}$.

Revision Tips

- Hard water contains $\text{Ca}^{2+}/\text{Mg}^{2+}$, prevents lather formation with soap.
- Soapless detergents use a hydrophilic head and hydrophobic tail.
- EDTA titration: 1:1 ratio with Ca^{2+} , titre $\div$ sample volume gives concentration.
- Colorimetry: higher absorbance = higher

concentration.

- Molecular polarity depends on shape and bond dipoles.
- Use bond enthalpies carefully: $\Delta H =$
bonds broken – bonds formed.